M.TECH · INFORMATION ARCHITECTURE

IA Midsem 2 — Units 3, 4 & 5

Organizing Information → Designing Navigation Systems → Labeling Systems.
Concise exam notes with tables, examples and only the diagrams that help.

Unit 3 · Organizing Information | Unit 4 · Navigation Systems | Unit 5 · Labeling Systems

Source: Information Architecture for the World Wide Web — Chapters 3, 4 and 5 · Covers only the supplied material

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How to use these notes

⭐ **Exam tip — Write every answer in this order**

1. **Definition** — one line (blue 📘 boxes).
2. **Key points** — 3–5 bullets with the **bold** keywords.
3. **Example or diagram** — and a table for “differentiate” questions.

Box	Use
📘 Definition	Opening line of an answer
⭐ Exam tip	Points that score marks
🧠 Yaad rakho	Short Hinglish memory hook
⚠️ Careful	Traps, problems, cautions
❓ Likely exam questions	End-of-unit practice list

Unit numbers follow the chapter numbers of the supplied material (Chapter 3 = Unit 3, and so on). Examples come from the source; nothing outside it has been added.

UNIT 3

Organizing Information

Challenges | Organization schemes | Exact vs ambiguous | Hybrid schemes | Hierarchy | Hypertext | Database model | Cohesive systems

3.1 Organizational challenges

📘 **Definition**

Classification = organizing information into categories so people can find and understand it. On the Web everyone now publishes, so everyone must label, organize and catalog — the job once done by librarians.



These four make organizing web content hard.

Fig. 1 — Four challenges of organizing information

1. Ambiguity

- Classification is built on **language**, and words have several meanings. *Pitch* has 15+ meanings (a throw, black sticky waterproofing, motion of a ship, a sales talk, sound frequency).
- Users may **miss the meaning** of a label.
- We must also agree **which documents go in which category**. Example: a **tomato** — fruit, vegetable or berry?
- Hardest for **abstract concepts** (subjects, topics, functions): e.g. *alternative healing* — philosophy, religion, health & medicine, or all?

2. Heterogeneity

- **Heterogeneous** = made of unlike parts; **homogeneous** = identical parts (Oreo cookies).
- Library card catalog = homogeneous: books only, each record has author, title, subject → structured scheme works.
- Web sites are heterogeneous in **two ways**:

Type	Meaning
Granularity	Items at different levels side by side — articles, journals, databases, sections, pages, other sites
Multiple formats	Text, images, video, audio, software, news, HR pages; each format has its own key facts (image: file format GIF/TIFF, resolution)

So a one-size-fits-all, highly structured scheme fails.

3. Differences in perspectives

- Labels and organization are shaped by the **creator's view** (e.g. files in `/office/clients/red` vs `/blue` make sense only to the owner).
- Companies organize by their **internal divisions** (marketing, sales, support, HR, IS) — a customer looking for technical help may not know where to go.
- Put yourself in the **intended user's shoes**; users differ in familiarity. A perfect system is impossible — understand the audience to do better.

4. Internal politics

- Every organization has politics; the choice of organization and labels affects how users see the company, its departments and products.
- Example: should the library get a main-page link? Call it *The Library*, *Information Services* or *Knowledge Management*? Then why not corporate communications or daily news?
- The architect must be **sensitive**, remind colleagues to design for the **user**, and sometimes **compromise**.

🧠 **Yaad rakho**

Challenges = **A-H-P-P**: Ambiguity, Heterogeneity, Perspectives, Politics.

3.2 Organizing web sites and intranets

📘 **Definition**

An **organization system** = **organization schemes** + **organization structures**.
Scheme — defines the *shared characteristics* of content items and the logical grouping.
Structure — defines the *types of relationships* between content items and groups.

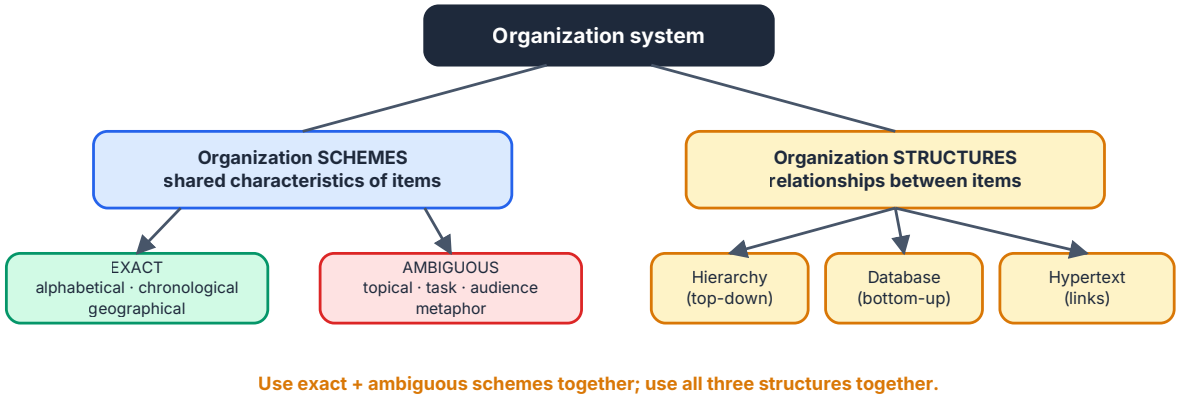


Fig. 2 — Organization system = schemes + structures

- Organization is closely tied to **navigation, labeling and indexing**; the hierarchy is often the primary navigation system and category labels define its contents.
- Focusing only on logical grouping avoids implementation distractions.

3.2.1 Organization schemes

Examples: phone book, supermarket, TV guide. Phone book white pages = **exact**; supermarket (marshmallows: snack aisle or baking?) = hybrid topical/task, **ambiguous**.

	Exact schemes	Ambiguous schemes
Sections	well defined, mutually exclusive	categories defy exact definition
Search type	known-item (you know the name)	browsing, associative learning
Design / upkeep	easy — little intellectual work	hard — needs classification + indexing
Examples	phone-book white pages, author/title	yellow pages, Dewey, subject catalog
Weakness	must know exact name	subjective, language is ambiguous

Fig. 3 — Exact vs ambiguous schemes

A. Exact organization schemes

- Divide information into **well-defined, mutually exclusive** sections; no ambiguity.
- Used for **known-item searching** (you know the name: *Porter* is in P, after O, before Q).
- **Problem:** user must know the exact name — white pages don't help to find a plumber.
- Easy to design and maintain; easy to use.

Scheme	Use / examples	Key points
Alphabetical	Encyclopedias, dictionaries, nonfiction indexes, phone books, store directories, bookstores, libraries (26 letters)	Primary scheme for reference works; often an **umbrella** for other schemes; sort by last name, product/service, department, format. Supports scanning + casual browsing (e.g. directory "Browse by Last Name")
Chronological	Press-release archives, history books, magazine archives, diaries, TV guides	Easy when everyone agrees when an event occurred. Users may also want title/keyword → a **combination** of schemes is often needed
Geographical	News and weather by place; political, social, economic issues; travel sites with a clickable map	Place matters to users; straightforward except **border disputes**

B. Ambiguous organization schemes

- Divide information into categories that **defy exact definition**; affected by language ambiguity and human **subjectivity** (the tomato problem).
- **Hard** to design, maintain and use — but often **more important and useful**.
- Library example: author and title are exact; **subject** (Dewey Decimal, Library of Congress) is ambiguous but used far more.

💡 **Why it matters — Why ambiguous schemes are useful**

- We **don't always know what we're looking for** or the correct label; the information need may be vague.
- Seeking is **iterative and interactive**; what you find shapes what you look for next → **associative learning** and serendipity ("web surfing at its best").
- Items are grouped in **intellectually meaningful ways** (alphabetical groups share only a letter).

- Success depends on a good **classification system** (hierarchy of categories and subcategories with **scope notes**) and careful ongoing **indexing** — assigning each item to categories accurately and consistently.

Scheme	Key points
Topical (subject)	Hard but very useful: yellow pages, academic departments, newspapers, book chapters. Few sites should be *only* topical, but most need some topical access. Define the **breadth of coverage** (whole human knowledge vs company products only) — you define the universe of content. Show **scope notes** for categories
Task-oriented	Content/applications grouped into **processes, functions or tasks** (word-processor menus: *Edit, Insert, Format*). Works when a **limited number of high-priority tasks** can be anticipated. Less common on the Web (sites are content-heavy) but suits **intranets/extranets**. Example: GM's "select a car, locate a dealer, choose a finance plan..."
Audience-specific	For **two or more clearly definable audiences**; best for **repeat visitors** who bookmark their section; allows customized, clutter-free **mini-sites**. **Open** = users may enter other audiences' areas; **Closed** = users cannot move between them (subscription fees, security). Example: SIGGRAPH 97 — contributors, exhibitors, media
Metaphor-driven	Relate the unfamiliar to the familiar (desktop, folders, trash can). Good for brainstorming. **Cautions:** (1) metaphor must be **familiar** to users (a hardware vendor organized like a motherboard fails); (2) it brings **baggage and limits** — a virtual library implies reference staff; services with no real-world match break the metaphor. Example: Bianca's "shack with rooms" — fun but not intuitive

🧠 **Yaad rakho**

Ambiguous = **T-T-A-M** (Topical, Task, Audience, Metaphor). Exact = **A-C-G** (Alphabet, Chronology, Geography).

C. Hybrid schemes

📘 **Definition**

A **hybrid scheme** blends elements of several pure schemes in one list. Confusion is almost guaranteed because users cannot form a single **mental model**.

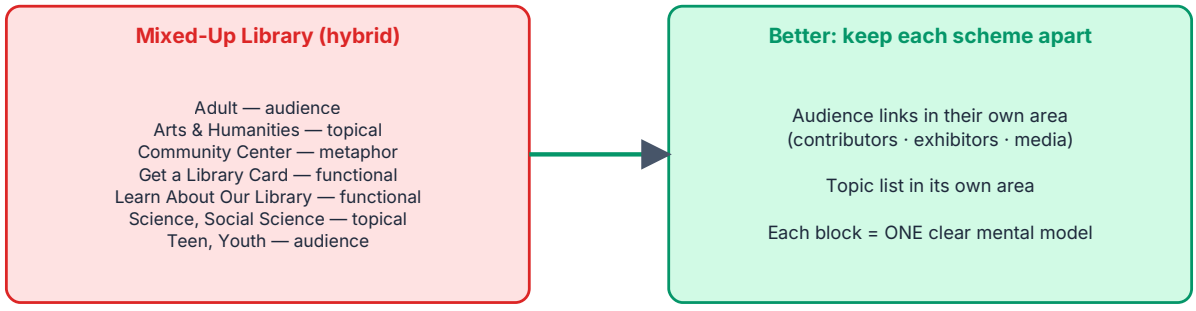


Fig. 4 — The "Mixed-Up Library" menu and the fix

- Common on the Web because teams cannot agree on one scheme for the main page.
- **Solution:** if multiple schemes must share a page, keep each **pure and presented separately** (e.g. SIGGRAPH page shows the audience scheme apart from the others).

3.2.2 Organization structures

Structure defines the **primary ways users navigate**. (Movies are linear; plots may be non-linear; maps are spatial.) Three major structures: **hierarchy, database-oriented model, hypertext** — each has strengths and weaknesses; often use **all three** together.

A. Hierarchy — top-down approach

📘 Definition

A **hierarchy** is a tree of **parent–child** subdivisions. It is the foundation of almost all good information architectures.

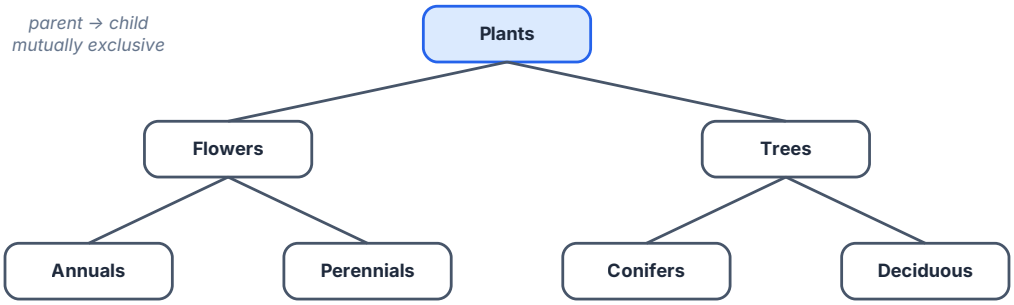


Fig. 5 — A simple hierarchy: Plants → Flowers / Trees → ...

- **Familiar:** family trees, kingdoms/classes/species, org charts, chapters → sections → paragraphs.
- Users quickly build a **mental model** of the structure and their location → comfort and context.
- Good starting point: gives the **scope** of the site fast, **without an extensive content inventory**; identify major content areas first, then schemes.

Designing hierarchies — rules of thumb

1. **Mutual exclusivity vs inclusivity.** Categories should be mutually exclusive, but ambiguous items (tomato: fruit or vegetable?) may be **cross-listed** in two or more categories so users find them. **Too many cross-listings destroy the hierarchy's value.** Different schemes may list the same items (topic vs format).
2. **Breadth vs depth. Breadth** = options per level; **depth** = number of levels.

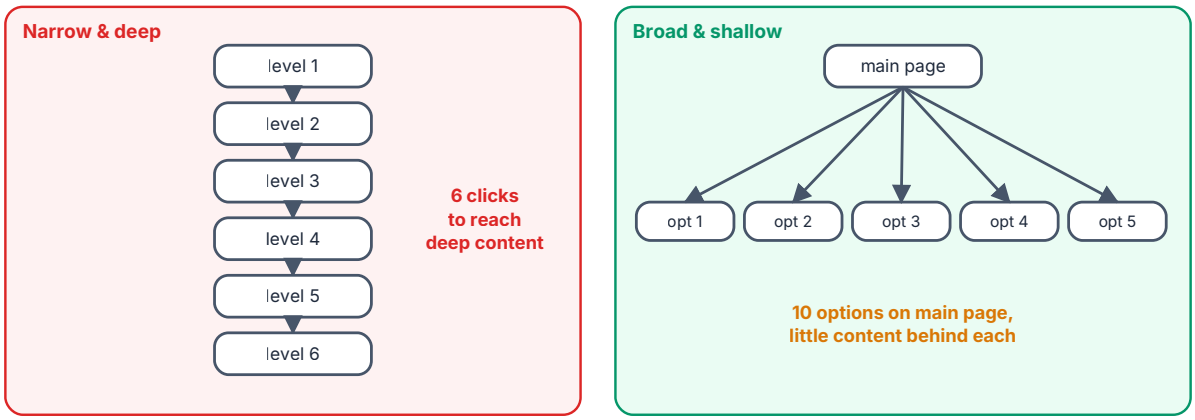


Fig. 6 — Narrow-and-deep (6 clicks) vs broad-and-shallow (10 options)

	Problem	Guideline
Too broad & shallow	Too many choices on the main menu; little content after a click	Follow Miller's **7 ± 2 rule**; more than ~10 options overwhelms users
Too narrow & deep	Inordinate number of clicks	Be more conservative: users go no deeper than **4–5 levels** or they give up

- **Growing sites:** lean **broad and shallow** — new items can be added without restructuring. Changes to the **main page** hurt the users' mental model and are **costly** (graphic design and layout get much care and money).
- Don't be trapped by the hierarchy — it is only **one component** of a cohesive organization system.

B. Hypertext

📘 **Definition**

Hypertext is a non-linear way of structuring information: **chunks** of information (text, data, image, video, audio) joined by **links**. Chunks can be linked hierarchically, non-hierarchically, or both.

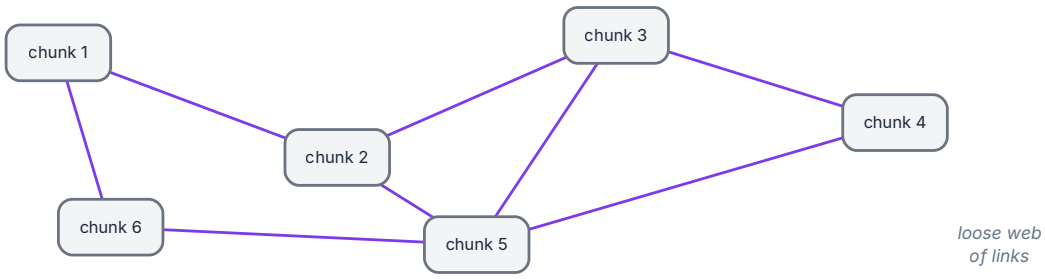


Fig. 7 — Hypertext: chunks connected in a loose web

- **Pro:** great flexibility; allows useful, creative relationships between items and areas.
- **Con:** high potential for complexity and confusion — users get lost (“bouncing from tree to tree in a forest”), cannot build a mental model, get overwhelmed. Links are often **personal**: one person's relationship may not be obvious to others.
- So hypertext is **rarely a good primary structure**; use it to **complement** hierarchy or database. Usually **design the hierarchy first**, then add hypertext links.

C. Relational database model — bottom-up approach

📘 **Definition**

A **database** is a collection of **records**; each record has **fields** (e.g. customer name, address, city, ZIP, phone).

	Points
Advantages	• **Field-specific searching** (one customer, or all users in a ZIP code) • **Easier content management**: global search-and-replace, data validation, security, **distributed management** with version control • **Repurpose** the same content in many forms/formats for different audiences (web, printed brochure, CD-ROM) — no separate HTML version per audience
Limitations	• Records follow **rigid rules**: same fields in each record type, consistent formatting in each field • Poor fit for **heterogeneous** content; moving all text, graphics and links of every HTML page into a database is technically hard, **expensive and slow**
Best for	Subsites/collections of **structured, homogeneous** information: staff directories, news-release archives, product catalogs

Designing databases

- Top-down hierarchy design reveals content areas suited to a database; involve a **programmer** for data-modeling issues.
- An **ERD (entity-relationship diagram)** visualizes and refines the data model **before** design and population. **Entities** (e.g. Resource) have **attributes** (Name, URL) → become **records** and **fields**. It shows relationships (e.g. each resource is at one or more locations; **1-to-many**).
- Then work **bottom-up**: identify content and structure of individual **record types** (e.g. staff directory: name, office phone required; email optional; *expertise* keywords — decide on a controlled vocabulary).

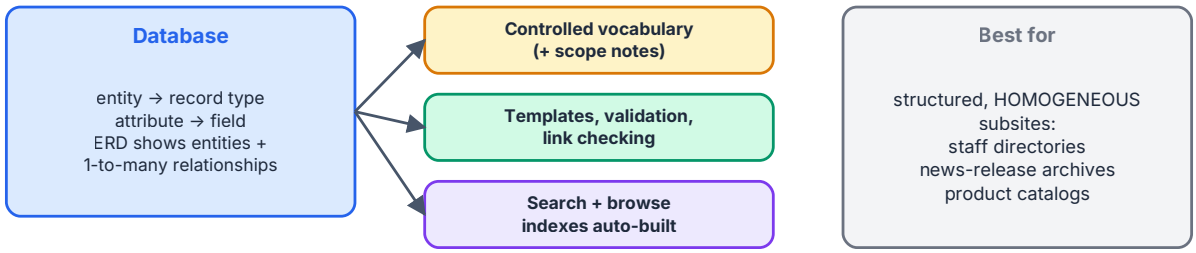


Fig. 8 — From database design to browsing and management

> 📘 **Definition**
>
> A **controlled vocabulary** specifies the **acceptable terms** for a field, optionally with **scope notes** defining each term. (Argus Clearinghouse ecology list: biodiversity, conservation, wildlife, wetlands conservation …; scope note: ecology = "the branch of biology dealing with the relation of living things to their environments".)

	Points
Benefits	Consistency in searching and browsing (search *biodiversity* — no need to try *biological diversity*); **browsable indexes generated automatically**; efficient to maintain
Drawback	Creating and maintaining one is **not simple** → combine a simple controlled vocabulary of broad categories with an **uncontrolled keyword field** (balance of structure and flexibility)

Searching, browsing and implementation

- Every field is another way to **browse or search**; browsing indexes and records are **generated from the database**; site-wide changes at the press of a button (e.g. Henry Ford Health System).
- More **efficient, accurate data entry**: admin interfaces hide HTML tags, **templates** ensure standard formatting, tools check syntax and links, indexes rebuilt automatically after every add/delete/change.
- **Implementation:** static HTML pages produced in **batch**, or dynamic pages generated **on-the-fly**. Chosen by technical performance (bandwidth, CPU) — little impact on the architecture.

Comparison of the three structures

Structure	Approach	Strengths	Weaknesses
Hierarchy	Top-down	Simple, familiar, mental model, quick scope	Limited; mutual-exclusivity vs cross-listing; breadth–depth balance
Database	Bottom-up	Field search, easy management, repurposing	Rigid; poor for heterogeneous content; costly
Hypertext	Non-linear	Flexible, creative links	Users get lost; personal links

3.3 Creating cohesive organization systems

- Organization systems are complex → **break the site into narrow domains** and tackle one question at a time (retrieval works best on **homogeneous** content).
- Don't lose the **big picture**: like cooking, good ingredients don't guarantee they mix well.

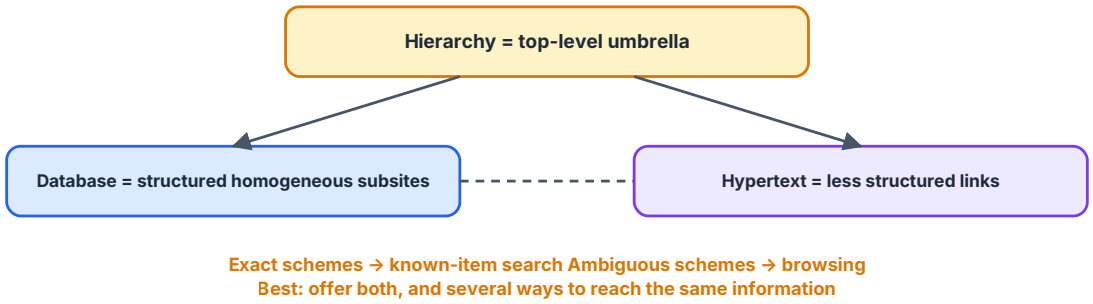


Fig. 9 — Combining schemes and structures

Guideline	Detail
Exact vs ambiguous	Exact → **known-item** search; ambiguous → **browsing / associative learning**. Whenever possible use **both**
Challenges	Language is ambiguous, content is heterogeneous, people differ, politics → provide **multiple ways** to reach the same information
Structures	Large sites need **all three**: top-level umbrella = **hierarchy**; structured homogeneous subsites = **database**; less structured relationships = **hypertext**

> ⭐ **Exam tip**
>
> Common 10-mark question: **"Explain organization schemes and structures."** Answer = system → exact (3) → ambiguous (4) → hybrid problem → hierarchy / database / hypertext → combine all three.

❓ Likely exam questions — Unit 3

- Explain the four challenges of organizing information.
- Define organization scheme and structure.
- Differentiate exact and ambiguous schemes with examples.
- Explain alphabetical, chronological and geographical schemes.
- Explain topical, task-oriented, audience-specific, metaphor-driven schemes.
- What is a hybrid scheme? Why is it a problem and how is it solved?
- Explain hierarchy design: exclusivity, breadth vs depth.
- Advantages and limitations of hypertext.
- Explain the database model; role of a controlled vocabulary and ERD.
- How do the three structures combine into a cohesive system?

UNIT 4

Designing Navigation Systems

Browser features · Context · Flexibility · Navigation types · Bars, frames, menus · Remote elements · Elegant design

Introduction

- Getting lost is bad (confusion, frustration, anger). **Navigation tools** give **context** and **flexibility of movement**.
- A good hierarchy reduces getting lost, but a **complementary navigation system** is usually needed.
- Navigation can support **associative learning** with *see also* links to related products/services (also helps marketing).
- **Challenge:** balance flexibility with the danger of **too many options**.

Element kind	Examples
On the content pages themselves	Graphical navigation bars, pop-up menus
Remote access to content	Tables of contents, site maps

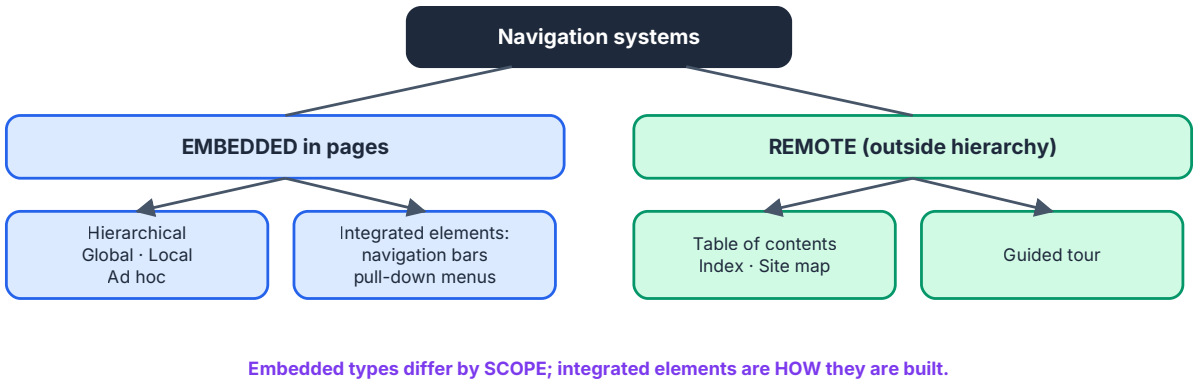


Fig. 10 — Navigation systems at a glance

4.1 Browser navigation features

Feature	Function
Open URL	Direct access to any page
Back / Forward	Bidirectional backtracking
History	Random access to pages visited in the session
Bookmark	Save the location of pages for later
Link colors ("bread crumbs")	Unvisited vs visited links in different colors → shows where you have been
Prospective view	Pointing at a link shows the destination **URL** at the bottom of the window — a hint about the content or an off-site destination

⚠️ **Careful — Do not override or corrupt the browser**

- **Link colors:** designers change them for aesthetics (e.g. match the logo) — a **sacrifice of usability**. Default colors help because users need not relearn the site. Modify only with an intelligent balance (keep the red/purple visited vs unvisited standard).
- **Server-side image maps** disable prospective view (shows XY coordinates instead of the URL). Use **client-side maps** or tables for graphical navigation bars.
- Leverage the browser's features; **web pages do not exist without a browser**.

4.2 Building context

- Like a **"You Are Here"** mark on a map: before moving you must know your position.
- The Web has **no natural landmarks**; hypertext, search results and printed pages let users skip the front door and land in the middle of a site → loss of context.

Rules of thumb

1. **Every page shows the organization's name** (in title or header) with a consistent graphic identity; a user entering through a subsidiary page must know which site and which area they are in.
2. **Navigation shows the hierarchy structure and the current location** in a clear, consistent way (e.g. the path Main Page → Government & Law → intellectual property). This builds a mental model.

4.3 Improving flexibility

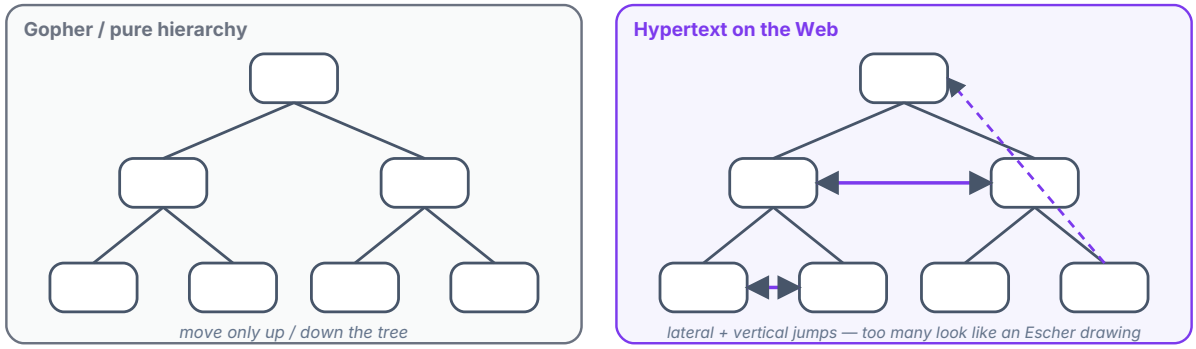


Fig. 11 — Gopher allowed only up/down; hypertext allows lateral + vertical movement

- Hierarchy is the foundation but **limiting**: in **Gopher** you could only move up or down the tree, not jump across branches (lateral) or between levels.
- **Hypertext** supports lateral and vertical navigation: e.g. jump from Programs & Events straight to Registration without retracing steps. It can **bypass the hierarchy**.
- Also allow vertical moves up a branch (specific program → Programs page) or back to the **main page**.
- **Trick:** balance flexibility against clutter. No lateral/vertical aids = limiting; too many = buries the hierarchy and overwhelms. Navigation should **complement and reinforce** the hierarchy with context and flexibility.

4.4 Types of navigation systems

Type	Scope	Key points
Hierarchical	Primary system	Main page → destination pages; options come directly from the hierarchy. Important but **limited** → needs other systems
Global	Whole site	Adds vertical + lateral movement everywhere
Local	One sub-site	Options unique to that area
Ad hoc	Single content items	Editorial links placed inside content

4.4.2 Global navigation

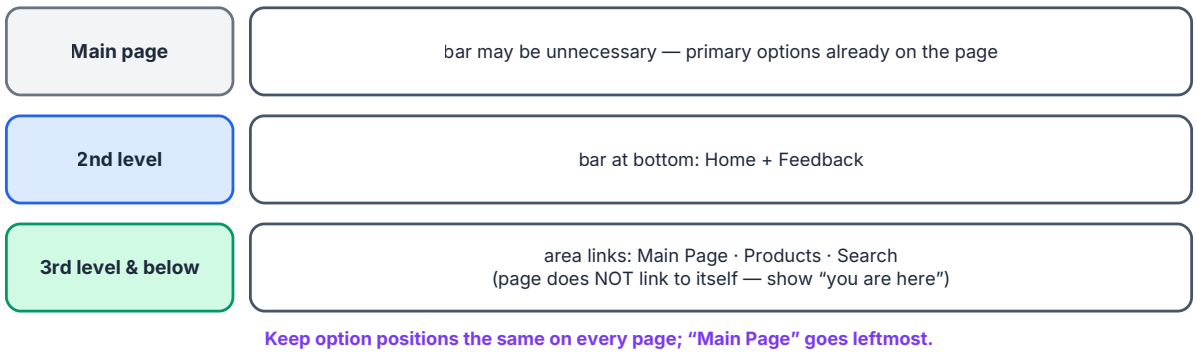


Fig. 12 — A rule-based global bar at each level

- Complements the hierarchy with **site-wide** vertical and lateral movement; built on **standard rules** per level.
- Simplest: a graphical bar at the **bottom of each page** (home page + feedback).

- Pages must **not link to themselves**; show "you are here" (folder-tab/button metaphor, or a logo symbol such as the **@ sign** on the Argus site).
- Integrate with graphic design; keep **relative positions the same**; since people read left to right, **Main Page goes leftmost**.

4.4.3 Local navigation

- Needed for complex sites. A **sub-site** (term coined by **Jakob Nielsen**) = pages within a larger site sharing a common style and navigation (e.g. an online product catalog: browse by name, format or market).
- **Extend the global system** through the sub-site (jump to main page, give feedback). Local **complements, not replaces**, global.
- Challenge: too many global + local options overwhelm the user → reduce the number or improve page design.

4.4.4 Ad hoc navigation

- Relationships that do not fit hierarchical/global/local; more **editorial than architectural** — an editor places links after content is in place.
- Usually **embedded links** in prose. **Problem:** usability tests show a **negative correlation** between embedded links and finding information — users scan fast and miss them.
- **Solutions:** give links their own lines, or a separate ad hoc menu at the top/bottom of the page; use embedded links only for **non-critical** "point of interest" links.
- The linked phrase must carry **context** (a bare "1861" gives no clue). **Moderation** is the rule — too many create clutter. **"Context is king."**

4.5 Integrated navigation elements

Elements built into content pages; most are **navigation bars** or **pull-down menus**.

4.5.1 Navigation bars

- Simplest form: hypertext links grouped together; or graphical (image map / images in a table).

	Text bar	Graphic bar
Look	Plain	Nicer
Speed	Fast	Slows loading (cached if reused across the site)
Maintenance	Easy to add/change options; **cheaper**	More work and cost
Users	Good for low bandwidth and text-only browsers	Use **ALT** text for replacement text

- Tip: graphic bar for **global**, text menu for **local** (use text if growth/change is expected).
- **Placement:** top and/or bottom, not the side (usability study: top/bottom did slightly better). Top = immediate access and context; bottom = after the page is read; choose by page length.
- **Labels vs icons:** textual labels are easiest and clearest; icons are hard to create and weak for **abstract concepts**, but help repeat users select quickly — use icons to **complement** text. **Pitfalls:** cultural differences in meaning; difficult to add options; color systems can help.

4.5.2 Frames

> 📘 **Definition**
>
> **Frames** split a browser window into independently scrollable **panes**; links in one pane control another. A static navigation bar stays visible while content scrolls.

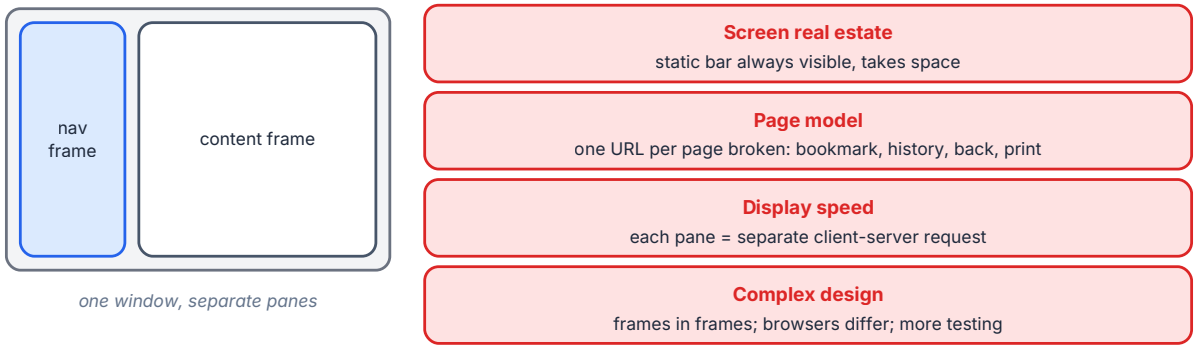


Fig. 13 — Frames give context but bring four problems

Problem	Detail
Screen real estate	Static bar always takes space (worse with blinking banner ads; on 640×480 screens). Keep it small; prefer a **vertical** frame (left-to-right reading suits narrow columns)
Page model	Each page has a unique URL; frames slice it up and often disable **bookmarking, visited/unvisited links, history**; confuse back, reload, print
Display speed	Each pane is a separate file → separate client–server interaction; slower, worse with heavy graphics
Complex design	Frames within frames when coming from another framed document; browsers/platforms display differently → more testing and overhead

4.5.3 Pull-down menus

- Show many options compactly in one line; usually standard **form syntax** — choose an option, then click **Go/Submit**.
- **Pop-up menus** (Java/JavaScript) appear on mouse-over of a word or area; the user picks directly.
- **Use with caution:** they let designers cram many options on a page, and **hide** options until the user acts. Work well for a **straightforward, exact** scheme (e.g. a location list).

4.6 Remote navigation elements

> 📘 **Definition**
>
> **Remote navigation elements** (supplemental systems: table of contents, index, site map, guided tour) are **external to the basic hierarchy** and give a **bird's-eye view** of the site, often in a separate window.

They are like documentation/help: **complement** a solid internal system, never replace or patch a poor one. **Provide them, but never rely on them.**

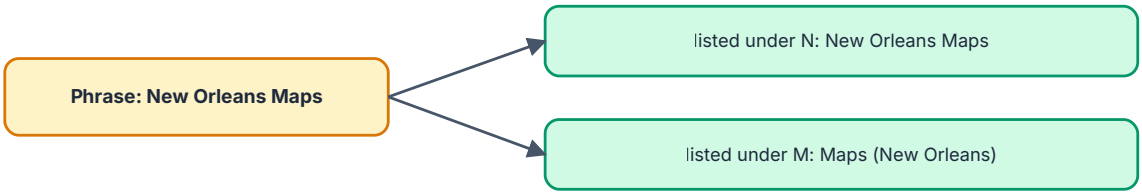
4.6.1 Table of contents

- Shows the **top few levels** of the hierarchy; supports random *and* linear access; hypertext links give direct access.

- Use for **hierarchical** sites; unnecessary for small sites of 2–3 levels; weak if the hierarchy is not strong.
- **Design rules:** (1) reinforce the hierarchy; (2) give fast, direct access to those who know what they want; (3) don't overwhelm — help, not scare.
- Graphics improve looks but cost more and slow loading — **function over form**.

4.6.2 Index

- Manually created **alphabetical keyword list**, **flat** (1–2 levels), without the hierarchy. Good for sites that are not strongly hierarchical and for **known-item** users.
- **Granularity** question: pages? paragraphs/concepts? collections? — ask *"What terms will users look for?"* Learn through **focus groups, interviews** before launch; **query tracking** after launch to refine.
- Index only **destination pages** (content), not **navigation pages** (menus) — the index lets users bypass navigation pages.
- **Term rotation (permutation)** rotates words so a phrase is found in two places. Apply selectively (not Sunday (Schedule) as well as Schedule (Sunday)).



Term rotation (permutation): the phrase is findable in two places. Use selectively.

Fig. 14 — Term rotation in an index

4.6.3 Site map

- **Site map** = a **graphical representation** of the site's architecture (excludes TOCs/indexes that are only decorated).
- Maps are traditionally for physical, not intellectual space. **Few good examples** because users are unfamiliar, designers are unfamiliar, and most sites have **multiple schemes and structures** (hypertext relationships are hard to show).
- Physical vs **symbolic** representation: higher abstraction → more intuitive. Don't show every link — only the **important** ones, clearly.

4.6.4 Guided tour

- Introduces **new users** to the main content areas; important for restricted-access/subscription sites (show what customers get for their money).
- **Linear** navigation (new users want guidance) plus a hypertext bar for flexibility; screenshots + narrative text.
- Also a **marketing** tool; few use it more than once → link it from the **gateway page** (splash page, audience-selection page, or preview page) rather than the main page.

4.7 Designing elegant navigation systems

- Focus on **context, flexibility and finding information** — not on flashy pop-ups or dynamic maps. **No single combination fits all**: consider goals, audience and content.
- **Process:** hierarchy → global → local → ad hoc (each higher level shapes the next) → then remote elements.
- Large, complex sites can use two or more remote elements (e.g. TOC + index), but watch confusion and maintenance cost.
- Build **usability testing** into the design process.

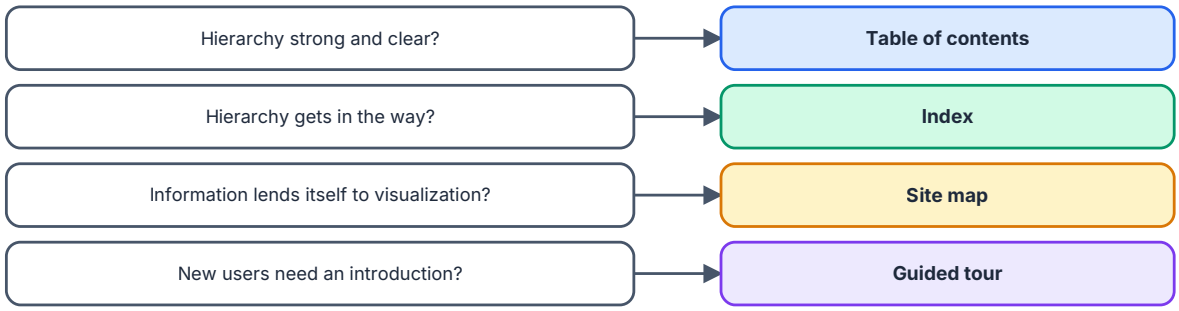


Fig. 15 — Which remote element? Ask these questions

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Order of design: **H-G-L-A** (Hierarchy, Global, Local, Ad hoc). Remote: **T-I-S-G** (TOC, Index, Site map, Guided tour).

❓ Likely exam questions — Unit 4

- Explain browser navigation features; how can designers corrupt them?
- How do you build context in a web site?
- Why is hierarchy alone not enough? Explain flexibility.
- Explain hierarchical, global, local and ad hoc navigation.
- Compare text vs graphic navigation bars; labels vs icons.
- What are the problems with frames?
- Pros and cons of pull-down menus.
- Explain table of contents, index, site map and guided tour.
- What is term rotation?
- Outline the process of designing elegant navigation.

UNIT 5

Labeling Systems

Why labels matter | Planned systems | Types of labels | Sources | Thesauri | Fine-tuning | Non-representational

Introduction

📘 **Definition**

Labeling is a form of **representation**: a label stands for a larger chunk of information (e.g. *Contact Us* = name, address, phone, fax, email). Goal: communicate information **efficiently** — without using up page space or the user's cognitive space.

- Labeling is **pre-recorded communication** (like print, scripted TV): no live feedback, so we must guess how users will respond — a great challenge.
- Labels are the most obvious way to show the organization and navigation systems; a page may mix label groups (hierarchy labels, site-wide navigation labels, sub-site labels). So decide organization and navigation **first**.

5.1 Why you should care about labeling

Reason	Explanation
Squandering attention spans	Users won't spend time decoding what labels like *Info* or *Stuff* really mean
Making bad impressions	Labels show what the organization is. Unprofessional words (*Cool, Hot, Stuff*) destroy confidence in a professional organization
Self-centered labeling	Org-speak labels are clear only to insiders (e.g. "Order Processing and Fulfillment Facility"; avoid *Total, Quality, Management*) → lose sales

5.2 Labeling systems, not labels

Labels form a **system** and need **planning**. Compare:

	Unplanned system	Planned system
Example	Faculty Skunkworks; Office for Instructional Technology; K12 PDN Projects Web Page; Project 1999; The New Media Center ...	Arts & Humanities; Business & Employment; Communication; Education; Engineering; Environment; Health & Medicine ...
Problem / strength	Mix of sites, units, programs, acronyms; first-time visitor can't tell what they represent	All **subject-oriented**, consistent
Completeness	Unclear	No exceptions → looks **comprehensive**
Learning	Each label must be decoded	User learns the **system** once → consistency breeds familiarity

5.3 Types of labeling systems

- Two formats: **textual** and **iconic**.
- Two uses: as **links** (navigation systems, index terms, link labels) and as **headings** that identify chunks on a page. One label can do double duty (link *Contact Us* → page titled *Contact Us*).

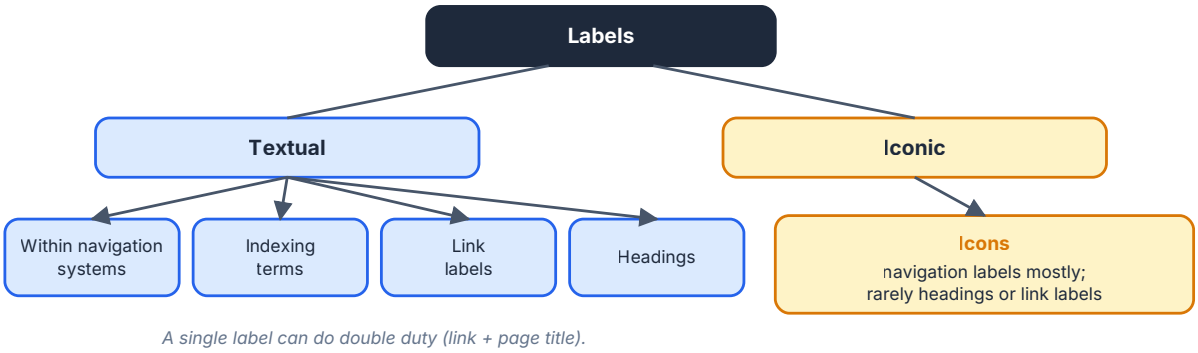


Fig. 16 — Types of labels

5.3.1 Labels within navigation systems

- Need **consistent application** more than any other labels — they repeat across the site. Don't call it *Main* on one page, *Main Page* on another, *Home* elsewhere.
- **Conventions** (use consistently within your site): Main/Home; Search/Find/Browse/Site Map/Contents/Index; Contact Us/Feedback; Help/FAQ; News/What's New; About Us/Who We Are.
- **Two problems:** several variants for one idea, and the **same label for different information** (News = new additions on one site, world events on another).
- **Fix: scope notes** — brief descriptions when first introduced, e.g.:

Label	Scope note
Search/Browse	Search by entering a query, or browse via a comprehensive site map
Contact Us	Direct line to customer service, 24-hour turnaround
News	Up-to-the-minute stock prices and press releases
Help	FAQ and how to contact the webmaster

5.3.2 Labels as indexing terms

- Used to classify content of large sites; two jobs: improve the chance of **retrieval by search**, and support **browsing**.
- **For searching:** keywords in the `<META>` tag or a database record — not usually seen by users; include services, locations, **synonyms and name variants** (e.g. *IFR Furniture Rentals*).
- `<TITLE>` tag terms: search-engine relevance ranking treats the title as very indicative of content → well-ranked results. Add accurate, descriptive terms.
- **For browsing:** manually added keywords give access **beyond the main organization scheme** (e.g. Henry Ford: heart, kidney, liver, lung… each keyword links to other content indexed under it).

5.3.3 Link labels

- Textual links within a chunk of text; easier to create because the **surrounding text gives context**. Out of context (Amalgamated, annual report, Bob Pobjoy…) they mean nothing.

- Consistency is not needed *between* link labels, but between a **label and the page it leads to** (link "Amalgamated" → page titled Acme Corporation confuses).
- Ask *"What kind of information will the user expect?"* before creating a link; be consistent (e.g. person names always lead to the same kind of destination, e.g. mailto).
- **Caution:** embedded links are hard to scan — fine for ad hoc links, not for frequently used navigation links.

5.3.4 Labels as headings

- Describe the chunk that follows; unlike link labels, there is **no guarantee the text is read**, so headings must work alone.
- Display all headings of the site as one **outline** and check two things:

Consistency in...	Meaning
Terminology	Wording is uniform and cohesive
Granularity	(1) chunks at each level are of roughly equal importance; (2) levels don't vary greatly in depth across the site

Example problems (GPSC site):

- Main page mixes **tasks** (Call our Telephone Hotline), **audiences** (For Prospective Employees) and **topics** (Publications for Sale) → caused by a poor organization system.
- Two publication pages use different forms: title with colon vs dash vs quotation marks; italics or not; "Ordering Information" vs "How to Order This Book"; different headings (synopsis, author) for no reason.
- **Granularity differs:** page 1 lists ordering by fax/telephone/Internet, page 2 only "How to Order" → lack of planning.

5.3.5 Iconic labeling systems

- Icons mostly serve as **navigation labels**; rarely headings or link labels.
- **Problem:** a much **more limited language** than text. Many icons exist for "home page"; an icon for "Press Releases" (newspaper? stacked pages?) is unclear. English has 600,000+ words with agreed meaning; icons have only about a **dozen** conventions.
- Study (Nielsen & Sano, Sun intranet): "Benefits" was read as *Clinton's health plan*, "What's New" as *Laundry*, "World Wide Web" as *dimensions of the planet*.
- Rely on **consistent positioning** on every page. Conventions: house = main page, question mark = help, magnifying glass = search. Few sites use icons **without text**.
- **Two reasons to use icons:** (1) consistent, attractive **graphic identity**; (2) familiar and easy to find — *if* drawn from a small set of known concepts and used consistently.

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>
> Labels as: **N-I-L-H** — Navigation, Indexing terms, Link labels, Headings (+ Iconic).

5.4 Creating effective labeling systems

Labels should mirror the **thinking and language of the users, not the owners**. The labeling system follows the organization system (topical organization → topical labels).

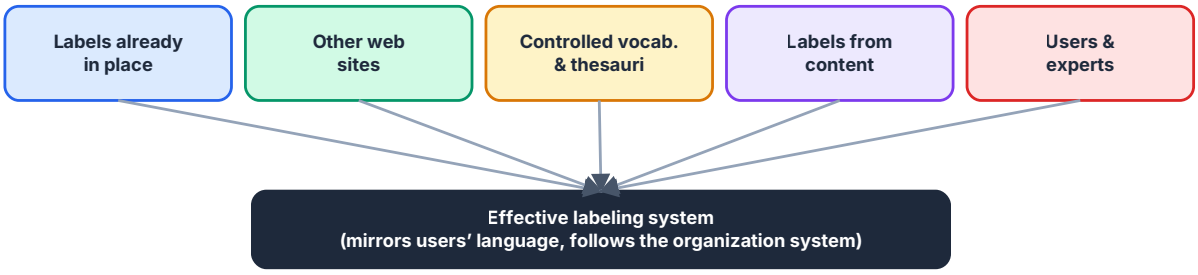


Fig. 17 — Five sources for labels

Source	How to use it
1. Labels currently in place	Starting point; capture all labels in a **label table** (page title graphic, `<TITLE>` tag, URL, headings). Condensed view exposes inconsistencies (What We Do / What We Do. / Web Site Design; Contact Argus / Contacting Argus)
2. Other web sites	"Borrow" good ideas; check competitors critically for your audiences' needs; academic sites (librarians)
3. Controlled vocabularies & thesauri	Pre-built professional lists (see below)
4. Labels from content	Use document titles/abstracts, but **authors aren't always the best labelers** — *Gone With the Wind* says nothing about its subject; authors may have artistic/marketing motives. Look at what is labeled; consider LCSH
5. Users and experts	Best source. **Query (search log) analysis** shows the terms users type, total searches and **zero-result queries** → add labels or content. Ask **experts** (librarians, switchboard operators) who know users' needs

Controlled vocabulary and thesaurus

📘 **Definition**

A **controlled vocabulary** = a list of **predetermined terms** describing a topic; you must use these terms, not alternatives (yellow-pages "Theatres-Cinema"). A **thesaurus** = a controlled vocabulary **plus relationships** between terms.

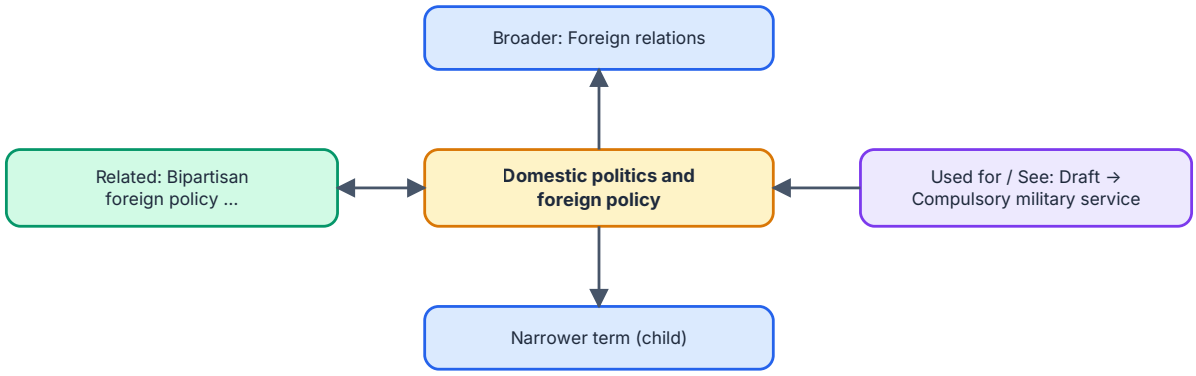


Fig. 18 — Thesaurus relationships (LIV example)

Relationship	Meaning / example
"See" / "Use"	Points from a term not in the vocabulary to the right one: *Draft* → use *Compulsory military service*
"See Also" / Related	Other terms of interest: *Domestic politics and foreign policy* ↔ Bipartisan foreign policy, Congress and foreign policy
Broader / Parent	Too specific? go up: Domestic politics and foreign policy → *Foreign relations*
Narrower / Child	More specificity: *Dog* is a narrower term of *Mammal*

- Library example: subject keywords come from the **Library of Congress Subject Headings (LCSH)**.
- **Narrower is better:** LCSH covers all human knowledge, is expensive to keep current and has arcane terms (*water closet*). Choose vocabularies focused on your audience, e.g. a computer-science thesaurus for computer scientists, or the **Legislative Indexing Vocabulary (LIV)** for legislative information.

Example: health-system labels from user needs

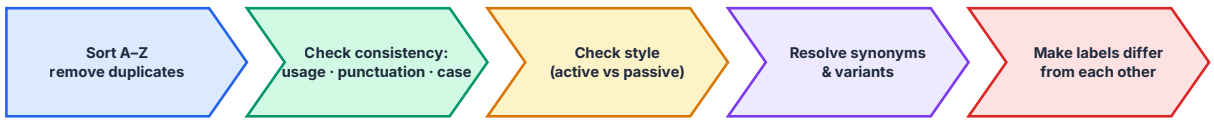
Medical-term labels were easy (existing thesauri); lay terms were not (no vocabulary, no content yet). Solution: list users' general needs, work with **librarians**, then label each.

Category	Sample labels
Problem / illness / condition	HIV, fracture, arthritis, depression
Organ / body part	heart, joints, mental health
Diagnostics / test	blood pressure, X-ray
Treatment / drug / solution	hospice, bifocals, joint replacement
Payment	administrative services, HMO, medical records
Health maintenance	exercise, vaccination

Use words the audience knows (e.g. *leg* instead of *femur*).

5.5 Fine-tuning the labeling system

5.5.1 The basics



Then fix scope (no phantom gaps) and size (right granularity).

Fig. 19 — Basic fine-tuning steps

- **Sort alphabetically** and remove duplicates.
- Check **consistency** of usage, punctuation, letter case (e.g. periods after labels; link vs heading labels).
- Check **writing style**: active verb ("Order a Free Sample…") vs passive ("…Customer Service"); set conventions.

- Handle **synonyms** (cancer / oncology), **variants** (microfiltration systems / services) and **related-but-not-same** terms (stationery / letterhead): pick the term that fits the users' language (medical audience → *oncology*); decide whether to merge or separate; specific vs broad.
- Make labels **descriptive and different** from one another (e.g. *global* and *international* side by side on the Fidelity site confused users).

5.5.2 Labeling system scope and size

Issue	Guideline
Gaps	Find obvious gaps; plan **phantom labels** for future content (online store with search but no ordering — will it add ordering?) to avoid cop-outs like *Miscellaneous, Other Info, Stuff*
Scope	Keep it **narrow and focused** on your unique content and audiences (LCSH lesson), yet comprehensive within that scope
Size	Navigation system: 5–10 terms. Large indexing system: hundreds. Choose the right **granularity**: 10 terms for 10,000 documents → long, unusable lists; hundreds of terms → users shy away. Find a **middle ground**; label **logical groups** (same department or author) instead of every document

5.6 Non-representational labeling systems

Labels need to be familiar and consistent — but **non-representational labels** are sometimes right. Not recommended regularly; hard to know when to apply. Two successes:

Case	Explanation
5.6.1 Good head-scratching	Confusion is normally bad, but it can create **mystery**. Personal-quest site whose labels are five generic words (where, I, searching, it, unfound) — they draw the user into a personal experience. Adding **scope notes** would destroy the mystery
5.6.2 When you just have to use icons	The Cool Central site: five "holes" with pictures that don't say much individually but together convey fun and invite exploration — works on a non-serious site meant for fun. e.g. sky/clouds → About; swimming fish → Nick's Picks; penguin → Site of the Moment; smoking detective → Nick Click, Private Eye

⭐ **Exam tip**

Exam answer for "labeling systems": **why care** (3) → **planned system** → **5 types** (navigation, indexing, link, heading, iconic) → **sources** → **fine-tuning** → scope/size.

❓ **Likely exam questions — Unit 5**

- Define labeling. Why should we care about labeling?
- Planned vs unplanned labeling systems.
- Explain labels within navigation systems and scope notes.
- Labels as indexing terms (META, TITLE tags).
- Link labels vs heading labels; consistency in terminology and granularity.
- Advantages and problems of iconic labels.
- Five sources of labels.
- Controlled vocabulary vs thesaurus; thesaurus relationships.
- Steps to fine-tune a labeling system; scope and size.
- What are non-representational labeling systems?

Quick comparison sheet

One-glance tables for last-minute revision.

Unit 3 — Organizing

Topic	Remember
Challenges	Ambiguity, Heterogeneity (granularity + formats), Perspectives, Politics
Exact schemes	Alphabetical, Chronological, Geographical — known-item search, easy
Ambiguous schemes	Topical, Task-oriented, Audience-specific (open/closed), Metaphor-driven — browsing, hard
Hybrid	Mixed schemes confuse → keep each scheme separate
Structures	Hierarchy (top-down), Database (bottom-up), Hypertext (links)
Hierarchy rules	Mutual exclusivity vs cross-listing; 7 ± 2; max 4–5 levels; growing site → broad & shallow
Database	Records/fields, ERD, controlled vocabulary; for homogeneous subsites

Unit 4 — Navigation

Topic	Remember
Browser features	Open URL, Back/Forward, History, Bookmark, link colors, prospective view
Context rules	Organization name on every page; show hierarchy + current location
Navigation types	Hierarchical · Global (site-wide) · Local (sub-site) · Ad hoc (editorial)
Integrated	Navigation bars (text vs graphic, top/bottom), frames (4 problems), pull-down menus
Remote	TOC (hierarchy), Index (flat, term rotation), Site map (graphical), Guided tour (new users)
Design process	Hierarchy → global → local → ad hoc → remote; test with users

Unit 5 — Labeling

Topic	Remember
Why care	Attention spans, bad impressions, self-centered labels
Types	Navigation · Indexing terms · Link labels · Headings · Iconic
Scope notes	Short descriptions that fix variants and same-label-different-meaning
Heading checks	Consistent terminology and granularity
Sources	Existing labels, other sites, vocabularies/thesauri, content, users/experts
Thesaurus links	See/Use · See Also · Broader · Narrower
Fine-tuning	Sort, consistency, style, synonyms, distinct labels; scope and size; no phantom gaps

Revision checklist

- ☐ Four challenges · scheme vs structure · exact vs ambiguous (with examples)
- ☐ Hierarchy rules (breadth/depth, 7 ± 2) · database + controlled vocabulary · hypertext limits
- ☐ Browser features and how designers break them · context rules
- ☐ Global / local / ad hoc · bars, frames, menus · TOC, index, site map, guided tour
- ☐ Label types · scope notes · iconic problems · sources · thesaurus relations · fine-tuning